

- 3 The Armand Bayou is a slow-flowing river that forms an aquatic ecosystem in Texas, North America.

Some biologists decided to study two invasive alien species of aquatic plants that are commonly found in the Armand Bayou: water hyacinth, *Eichhornia crassipes*, and malancha, *Alternanthera philoxeroides*. Water hyacinth and malancha reduce biodiversity in the Armand Bayou.

Water hyacinth is a floating plant that grows on the surface of the river. Water currents and the wind can move water hyacinth plants from one place to another.

Malancha has roots that attach the plant to the soil at the edge of the river. Malancha grows and reproduces to form dense mats on the river surface, which can trap floating water hyacinth plants.

Figure 3.1 shows water hyacinth and Figure 3.2 shows malancha.



Figure 3.1

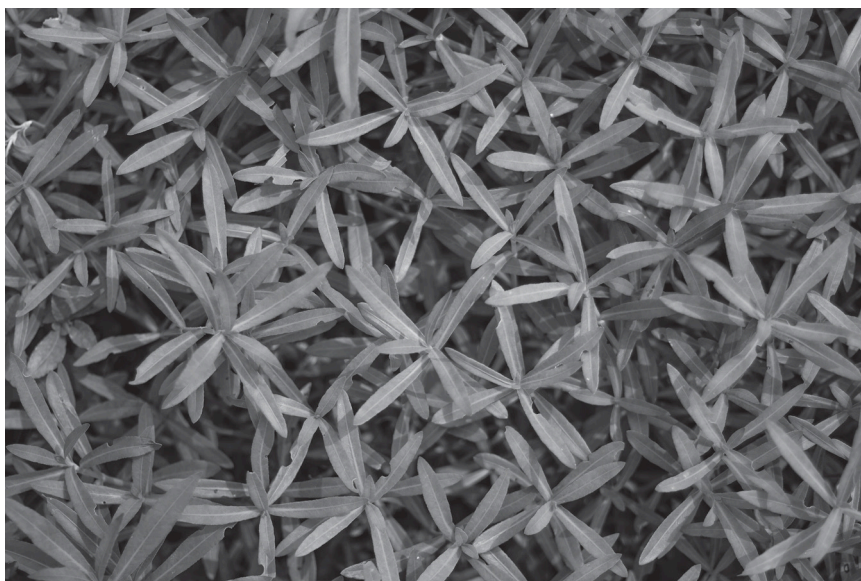


Figure 3.2

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

The biologists carried out a survey at 69 locations in the Armand Bayou. At each location, both invasive alien species were present.

- (a) The biologists estimated the percentage cover of water hyacinth and the percentage cover of malancha at each of the 69 locations. For these two variables, the biologists calculated Pearson's linear correlation coefficient, r , as $+0.389$, which was significant at $p < 0.001$.

Complete the sketch graph in Figure 3.3 to show the relationship between these two variables. Include axis labels in your answer.



Figure 3.3

[2]

- (b) The biologists carried out an investigation at 12 of the 69 locations in the Armand Bayou where both invasive alien species were present. The biologists wanted to test the hypothesis that:

The removal of malancha from the Armand Bayou will reduce the abundance of water hyacinth.

Figure 3.4 shows a map of the Armand Bayou and the surrounding area.

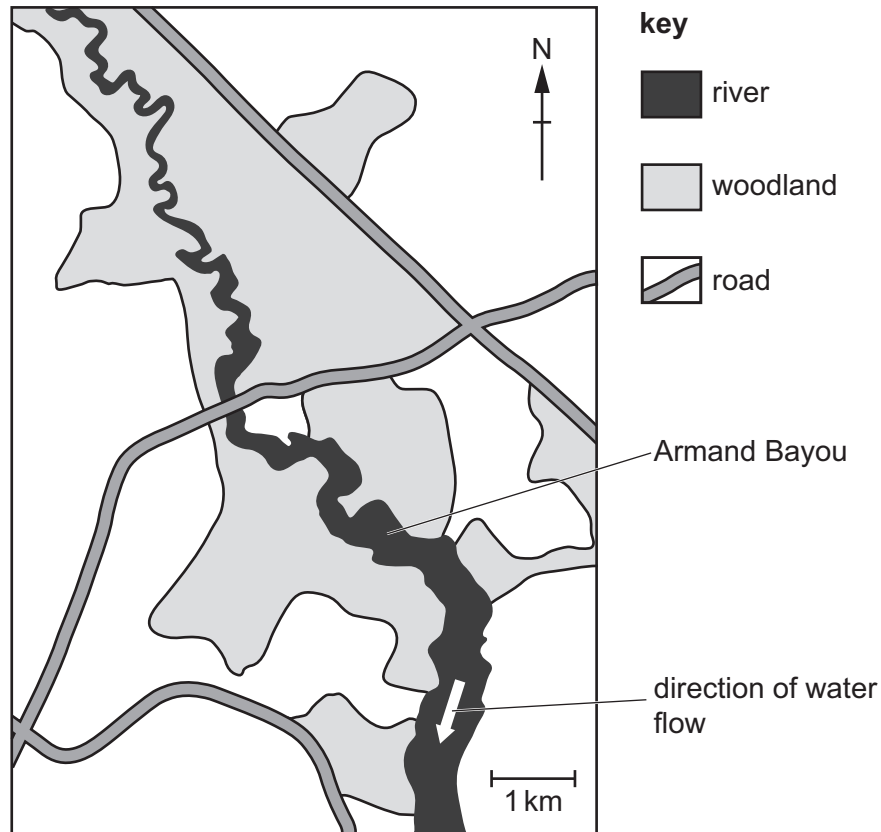


Figure 3.4

At each of the 12 locations, a 3 m × 5 m study area was marked out as shown in Figure 3.5.

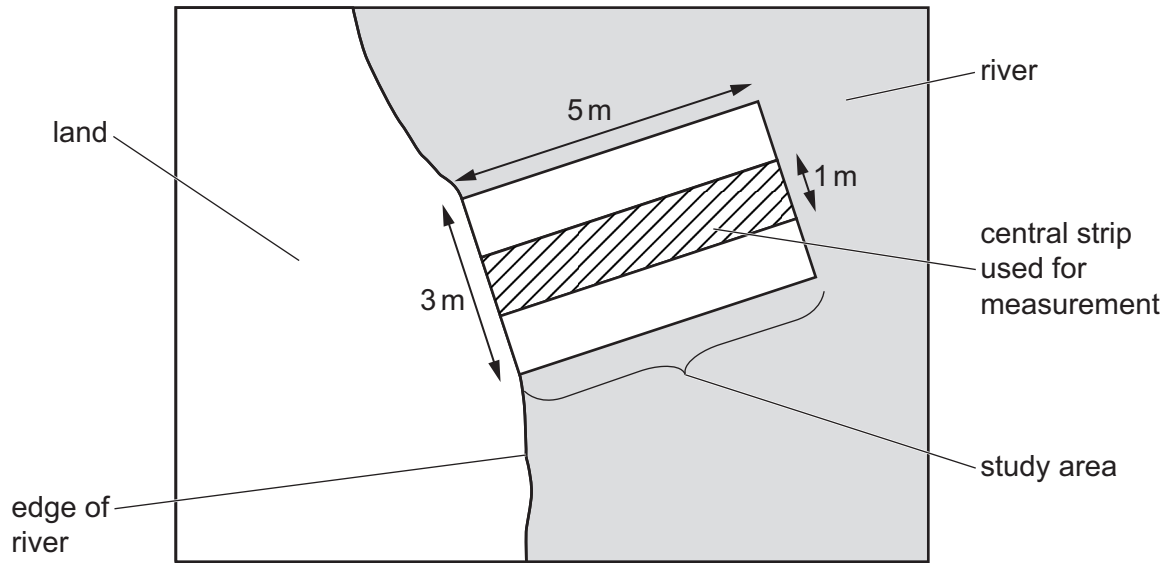


Figure 3.5

- (i) Describe a method that the biologists could use to select 12 locations at random from the 69 locations used in 3(a).

.....

.....

.....

.....

..... [1]

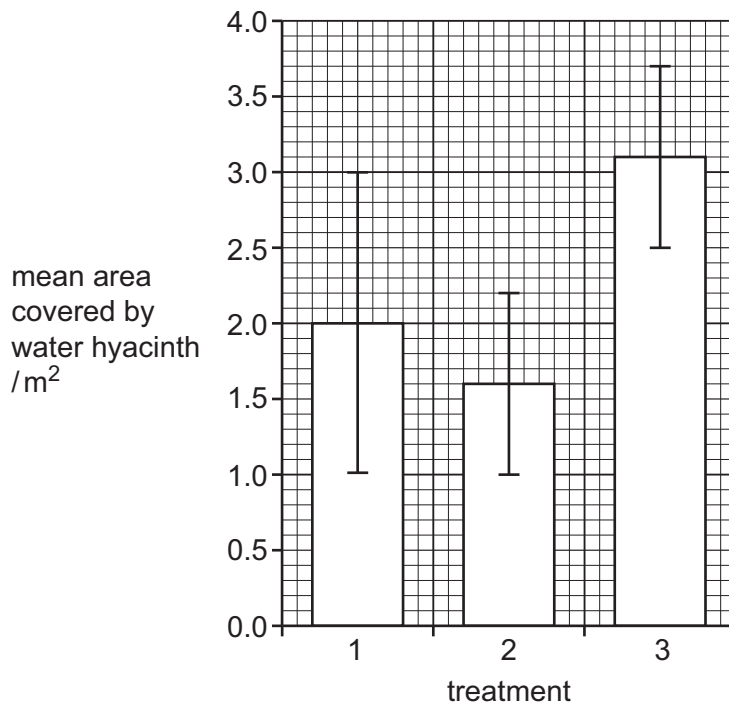


Figure 3.6

- (ii) The mean area covered by water hyacinth in locations given treatment **2** is smaller than the mean area covered by water hyacinth in locations given treatment **3** (control).

Use the data in Figure 3.6 to calculate this difference as a percentage of the results for treatment **3**.

space for working

percentage difference = [2]

- (iii) State which statistical test could be used to establish the statistical significance of this difference between locations given treatment **2** and locations given treatment **3**.

Explain why the test chosen would be suitable.

statistical test

explanation

.....

.....

- (iv) The risk assessment for this investigation stated that the biologists should wear gloves to prevent an allergic reaction to the different plant species.

Suggest **one other** hazard of the investigation, the risk associated with the hazard, and the precaution that would need to be taken.

hazard

.....

risk

.....

precaution

.....

[1]

- (v) A conservation volunteer wanted to reduce the abundance of invasive alien plant species found in the Armand Bayou.

The conservation volunteer analysed data from the investigation and concluded that removing malancha would reduce the abundance of water hyacinth in the Armand Bayou.

With reference to Figure 3.6 and other information provided about the investigation, evaluate the extent to which this conclusion can be supported.

.....

.....

.....

.....

.....

.....

.....

.....

.....

[4]

[Total: 11]